SecureWV

In partnership with



SWV Labs Threat Hunting with Wazuh

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Goal: This is the first of two Threat Hunting Labs! In this session we will be spinning up an Instance of the open source SIEM / XDR tool Wazuh and a client machine!

Pre Requirements:

- Personal laptop
- Preferable:
    - Download and install Virtual Box ahead of time.
    - Download and boot Wazuh OVA file ahead ahead of time will allow you to follow on from the start instead of having to download and import the VM.
    - At the time of writing this lab Wazuh requires just under 50 GB of disk space. You can substitute Kali if you want as many of the tools are pre-installed.
    - You may need to temporarily disable your local Firewall if you are having trouble.

“

Endpoint and Cloud Workload Protection

Wazuh unifies historically separate functions into a single agent and platform architecture. Protection is provided for public clouds, private clouds, and on-premises data centers.

Endpoint Security	Threat Intelligence	Security Operations	Cloud Security
• Configuration Assessment • Malware Detection • File Integrity Monitoring	• Threat Hunting • Log Data Analysis • Vulnerability Detection	• Incident Response • Regulatory Compliance • IT Hygiene	• Container Security • Posture Management • Workload Protection

Active XDR protection from modern threats
Wazuh provides analysts with real-time correlation and context. Its active responses are granular and include on-device remediation to keep endpoints clean and operational.

A comprehensive SIEM solution
The Wazuh Security Information and Event Management (SIEM) solution provides monitoring, detection, and alerting of security events and incidents.
“

https://wazuh.com/ (20251121)

Legal Disclaimer:

Before we go any further, it is critical to only use these tools on device you either own or have been given explicit (preferably written) authorization to perform these explicit acts on. The Computer Fraud and Abuse Act is a very real and serious thing. We all here at QCC want you to stay on the correct side of prison, the outside.

Prerequisite:

Download VirtualBox:

1. Go to: https://www.virtualbox.org/ and click downloads
2. Select your distribution to start your download
3. So much simpler than VMWare

Install Virtual Box:

1. Go to download location and double click. Accept UAC pop up.
2. Please note that exact install options might change as app is updated. This guide was created 8/24/2024.
3. On the first generic install screen click next.
4. The next screen allows you to customize your Virtual Box install. I kept with the default and clicked next.
5. Click Yes to the network interruption message.
6. Click yes if you get any missing dependencies.
7. At the next customization screen, I kept the default.
8. Click Install on the ‘Ready to Install Screen’
9. Click Finish

Make sure that you are connected to the internet.

Now we need to get the Wazuh free Virtual Appliance. Visit this site and click the free button.
https://documentation.wazuh.com/current/deployment-options/virtual-machine/virtual-machine.html
Download the linked virtual applicane (OVA)

Requirements: (at the time of the writing of this lab)

Version	CPU (cores)	RAM (GB)	Storage (GB)
Wazuh v4.14.1 OVA	4	8	50

OS Credentials
U: wazuh-user
P: wazuh
!
SSH root user login has been disabled. Wazuh-user has sudo privileges with 'sudo -i'

Web Dashboard:
U: admin
P: admin
URL: https://<WAZUH_SERVER_IP>
*find your ip with the command line command 'ip a'

Wazuh config files:

- Wazuh manager: /var/ossec/etc/ossec.conf
- Wazuh indexer: /etc/wazuh-indexer/opensearch.yml
- Filebeat-OSS: /etc/filebeat/filebeat.yml
- Wazuh dashboard:
    - /etc/wazuh-dashboard/opensearch_dashboards.yml
    - /usr/share/wazuh-dashboard/data/wazuh/config/wazuh.yml

All information here was gained from:
https://documentation.wazuh.com/current/deployment-options/virtual-machine/virtual-machine.html

For a real world deployment:

- We recommend using a traditional install and not the OVA. However for a home lab the OVA is just fine!
- We also recommend either assigning a static IP address or DHCP reservation for the Wazuh Server.

Wazuh Linux OVA into VirtualBox as the Hypervisor

Now we need to load the Wazuh Virtual Appliance into the Virtual Box:

1. As you will need you host system and yout Grenbone VM to have an IP address so that they can communicate with each other connect to a wifi (wither QCCLabs or your cell phone hotspot as in this case you will not be in in conflict with the Computer Abuse and Fraud Act)
2. Open Virtual Box
3. In the Virtual Box Console click 'File -> Import Appliance' or press 'Control + I'

4. Click the icon of the folder with an up arrow on the right of the import wizard and point to the OVA file you just downloaded. Click 'OK' to populate the File location. Once the File location is populated click 'Next'.
5. Review the settings.
    a. Double click on the name and change it to Wazuh if it does not auto populate.
    b. Double click on CPU to change the number of cores you give your VM. (min requirements at time of writing this is 4 cores)
    c. Double click on the ram number to allow you to change it to at least 5120 MB. (min per documentation)
    d. Adjust the 'Machine Base Folder' if you want you VM in a different place.
    e. Click Finish
    f. You might be asked to agree the Software License Agreement
6. You can watch the progress as Virtual Box imports the VM.
7. Once the Wazuh Virtual Appliance has been imported. Either double click the Wazuh VM or apply a single click to Wazuh VM and then click start. If you get an error like I did we need to do a few more steps.
    a. Click on the VM and then the golden Settings gear
    b. Click on the Networking icon in the Settings pop up.
    c. Ensure you in the Adapter 1 tab
    d. The official Wazuh instructions say you need to select the proper Networking Name under the 'Name' field. I had to go in and select the correct name of my network card to make it work.
8. Now try to power on your system again. I know apologize for the vagueness of 'play with the network settings until you resolve the issue with a virual appliance that someone else made'. But, we are hackers! Sometime we will relentless bash our head into a technical issue until we get good. It is like the Poet Laureate Chumbawamba once sang, "Oh, Danny Boy, Danny Boy, Danny Boy, I get knocked down, But I get up again, You're never gonna keep me down, I get knocked down, But I get up again, You're never gonna keep me down, I get knocked down, But I get up again, You're never gonna keep me down, I get knocked down, But I get up again, You're never gonna keep me down"

If you are using Virtual Box, we need to make adjustments to time config

1. In the VirtualBox console select the Wazuh virtual appliance with a left click, then right click and select 'settings'
2. Click on 'Settings' in the left side of the pop up window. Then in the 'Motherboard' tab check 'Enable Hardware Clock in UTC Time' and click 'OK'.
3. Re-enter the 'settings' for the Wuzah for the setting above. If the setting did not take, re apply. If it did click on the 'Network' icon on the left hand side of the config window and ensure 'Attached to' is set to 'Bridged Adaptor' and the 'Name' is your active network adaptor. If the network settings are not set this way, make it so!!! Sorry about the Star Trek reference, I am actually an Extended Universe Star Wars myself.
4. Click 'OK'

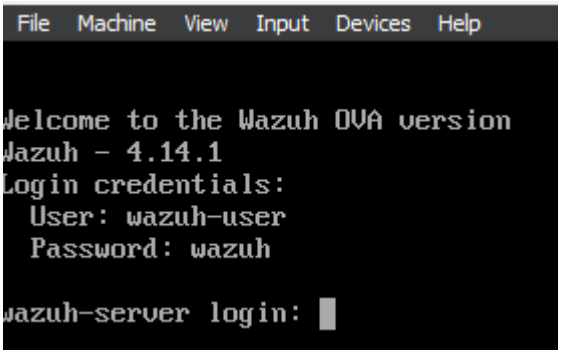
Initial Boot of Wazuh Virtual Appliance

Now star the Wuzuh VM in Virtual Box!

If you run into any challenges, you can follow the trouble shooting steps in:
https://documentation.wazuh.com/current/deployment-options/virtual-machine/virtual-machine.html

Please note that once you click into the Wuzah VM, your mouse might be capture by the Wuzah VM. To get you mouse back to the host OS type ‘Control + C’

A successful boot should look like:



Log in with provided OS credentials and get your ip:

OS Credentials

U: wazuh-user

P: wazuh

!

SSH root user login has been disabled. Wazuh-user has sudo privileges with ‘sudo -i’

Now power off the VM we just built.



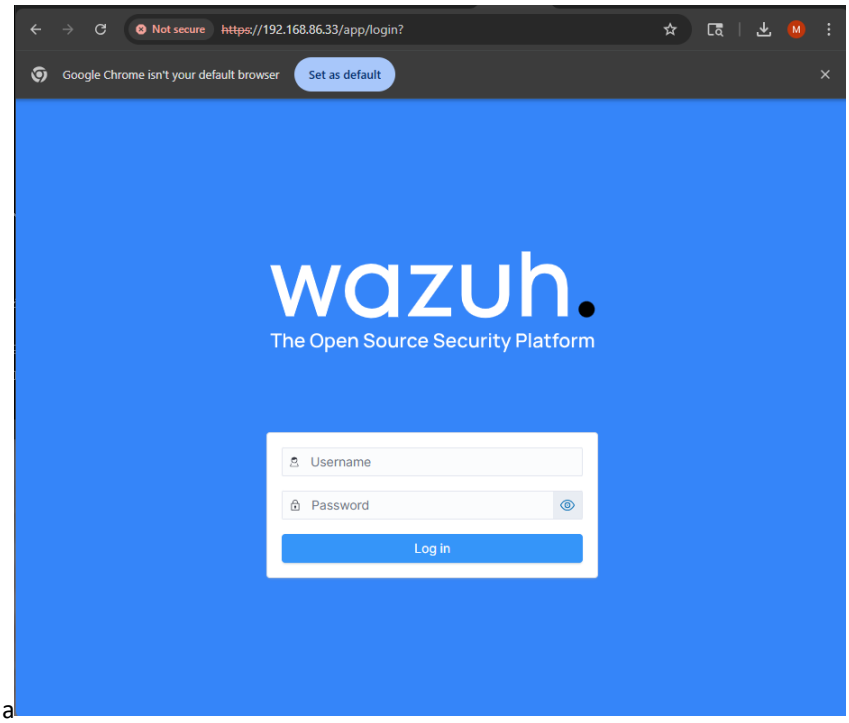
```
[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e9:a8:7e brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 192.168.86.33/24 metric 1024 brd 192.168.86.255 scope global dynamic eth0
       valid_lft 86212sec preferred_lft 86212sec
    inet6 fe80::a00:27ff:fee9:a87e/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$
```

You can see my IP is on the 192.168.86.0/24 subnet. Since I this VM is on the Bridged networking mode, this is the same network as the network my host system is on!!!

This means ANY VM we want to add to this instance MUST also use Bridged networking mode!

Now we move to your host system

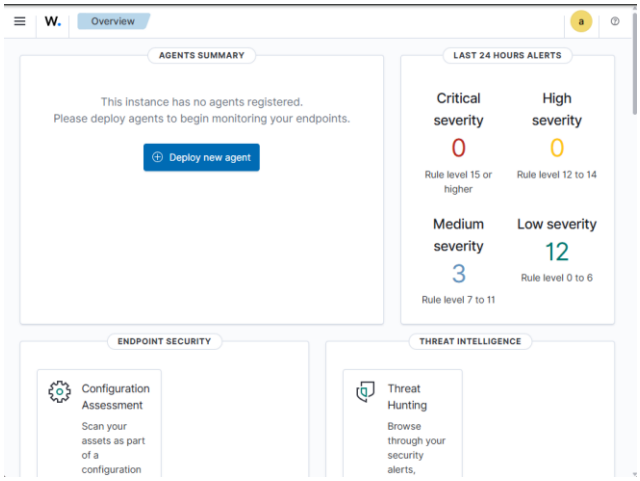
- Open the web browser of choice and navigate URL: https://<WAZUH_SERVER_IP> (where WAZUH_SERVER_IP is found by the output of the 'ip a' command above in my home lab this is URL: https://192.168.86.33)
- As the OVA uses a self-signed certificate you might have to accept the self-signed certificate (the exact actions differ from browser to browser but it will be along the line of 'Advanced' -> 'Proceed to [IP Address]'



a

- Yatta Yatta! (Click link at your own risk: https://www.youtube.com/watch?v=KUOwbcdZozQ&t=16s) YOU WUZUH IS WORKING
- Log in with the GUI console:
    - U: admin

- P: admin



Building the Wuzuh Client

You can expect your system build to take between 15 to 30 minutes power.

During the initial part of system setup of your VM you mouse might be captured only to the VM, in Virtual box to return your mouse to the host OS press 'Control + C'.

*When building Wazuh it might be faster to use 4096 ram and 4 CPU during the build process, once you shut down the VM you can always go into the Virtual Box Settings and edit the CPU to 2 / RAM 2048 to make it easier to run your system

1. Download ubuntu: https://ubuntu.com/download/desktop
2. Click new:
    a. Give OS Name
    b. Select desired location for VM files
    c. Select Ubuntu ISO
    d. Unclick skip unattended install. (Sometimes the old ways are best)
    e. Click Next
3. On the next screen select:
    a. ram 4096 (suggested build), 2048 (suggested run)
    b. 4 cpu (suggested build), 2 cpu (suggested run)
    c. Click next
4. Set hard drive size to 10 GB and press next
5. Click finish
6. Click on the newly created VM and click start
7. Select 'Try or Insta Ubuntu'
8. Choose your language and click next
9. At accessibility Click next

10. Select keyboard layout and click next
11. At ‘connect to internet’ select ‘use wired connection’ (#VirtualBoxMagic)
12. Click ‘Install Ubuntu’ and click next
13. Click ‘Interactive installation’ and click next
14. Click ‘Default selection’ and click next
15. At ‘Install recommended proprietary software’ leave all unchecked and click next
16. Select ‘Erase disk and install Ubuntu’ and click next
17. Fill out your username details and click next. Suggest something generic. I used:
    a. Set your host name to either **WazuhClient depending on which system you are building**
    b. U: ubuntu
    c. P: ubuntu
    d. (Yes a wrote this into the guide so that I would not forget as I am writing this on 12/4/2025)
    e. Click next
18. Select the zone of time and click next
19. FINALLY CLICK INSTALL!
20. Finally now we have nothing to do but wait! Might a suggest that you enjoy a responsible beverage of your choice!?!?!?
21. At the end use the Devices -> optical drive to remove the Ubuntu ISO from the CD mapping and reboot the machine.
22. Log into the system and click next on the Welcome to Ubuntu screen
23. Select ‘skip for new’ and click next
24. I selected the option that started with ‘No,’ and clicked next
25. Click finish
26. In the lower left hand corner click on the icon of the ring with three dots. Right click ‘terminal’ and drag and drop it onto the desired location on the quick launch bar on the left.

By default Virtual Box will put your VM in vlan local to your system and use NAT (Network Address Translation) in order to talk to systems on the outside. This is the same way your local router allows your system to communicate with the outside world. When you VM is turned off, you need to go into settings -> network -> Attached to and set it to Bridge. This will place your VMs on the same VLAN as your host machine.

Log onto you WazuhClient VM. Once logged in right click on the icon of a circle with three dots in the lower left hand corner, right click Terminal and select ‘Pin to Dash’.

Log onto the console of your WazuhServer.

Now we want to test our connectivity.

1. On your WazuhServer use the ‘ip addr’ command to get the ip address of the server. Note this in a text file.

```
[wazuh-user@wazuh-server ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e9:a8:7e brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 192.168.86.35/24 metric 1024 brd 192.168.86.255 scope global dynamic eth0
       valid_lft 86360sec preferred_lft 86360sec
    inet6 fe80::a00:27ff:fee9:a87e/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

2. Open the terminal in your WazuhClient and use the command ‘ip a’ to get your clients IP. Note this in your text file

```
ubuntu@wazuhclient:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f4:94:3f brd ff:ff:ff:ff:ff:ff
    inet 192.168.86.34/24 brd 192.168.86.255 scope global dynamic noprefixroute enp0s3
       valid_lft 80321sec preferred_lft 80321sec
    inet6 fe80::a00:27ff:fef4:943f/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@wazuhclient:~$
```

3. You can see that both VMs are in the same subnet (192.168.86.0/24). Now form each system ping the other to test connectivity with the ‘ping [destination ip or name]’. Use ‘Control + C’ to end the ping (you just need a handful of consecutive successful results.

```
[wazuh-user@wazuh-server ~]$ ping 192.168.86.34
PING 192.168.86.34 (192.168.86.34) 56(84) bytes of data.
64 bytes from 192.168.86.34: icmp_seq=1 ttl=64 time=0.508 ms
64 bytes from 192.168.86.34: icmp_seq=2 ttl=64 time=0.697 ms
64 bytes from 192.168.86.34: icmp_seq=3 ttl=64 time=0.472 ms
^C
--- 192.168.86.34 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2050ms
rtt min/avg/max/mdev = 0.472/0.559/0.697/0.098 ms
[wazuh-user@wazuh-server ~]$
```

```
ubuntu@wazuhclient:~$ ping 192.168.86.35
PING 192.168.86.35 (192.168.86.35) 56(84) bytes of data.
64 bytes from 192.168.86.35: icmp_seq=1 ttl=127 time=1.45 ms
64 bytes from 192.168.86.35: icmp_seq=2 ttl=127 time=0.427 ms
64 bytes from 192.168.86.35: icmp_seq=3 ttl=127 time=0.533 ms
64 bytes from 192.168.86.35: icmp_seq=4 ttl=127 time=0.288 ms
^C
--- 192.168.86.35 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3256ms
rtt min/avg/max/mdev = 0.288/0.673/1.447/0.454 ms
```

4. You can see we have success. If your systems are not on the same subnet with different IP address or your systems can not ping each other shutdown your VMs and double check your VM’s networking is set to Bridged.

Else raise your hand and the proctor or one of your table mates can help you troubleshoot. (Often we learn more in troubleshooting than in everything working smoothly. Do not proceed until your VMs can ping each other.

5. Ok So I am sure you can see the one weakness of this lab is we are relying on DHCP. This means that as we power down / power up / pause & un-pause our VMs, the DHCP Server may give out a different IP to our VMs. This is desired behavior for most systems and is ZERO issue to our WazuhClient (just as our normal desktops / laptops). However, this is a BIGLY deal for our WazuhServer. When you enroll clients to the WazuhServer, you will pass a special command you will see later that sets the WazuhServer's location on the network to either an IP, hostname, or Full Qualified Domain Name (FQDN). In the real world we advise configuring the WazuhServer with either a Static IP (Preferred) or a DHCP Reservation (Based on MAC address) to prevent the WazuhServer from changing IPs. While the goal is to give you a working lab to learn on, we recommend that if you want to continue to learn that you use what you have learned here and build a fresh instance at home with clients that are often in use so that you can see real world use cases.
6. To solve the WazuhServer floating IP we are going to set up a hostname of QCCWazuh.
    a. On your WazuhClient open up your terminal and type cd 'cd /etc/

```
ubuntu@wazuhclient:~$ cd /etc/
ubuntu@wazuhclient:/etc$
```

    The /etc directory in Linux is a fundamental part of the file system hierarchy, primarily serving as the central repository for system-wide configuration files and directories. The name /etc originally stood for "et cetera" in early Unix systems, signifying a place for miscellaneous system files. One the file we are interested in '/etc/hosts' which we can manually bind a hostname to an IP.

    b. Ping QCCWazuh and you will see we get a fail

```
ubuntu@wazuhclient:/etc$ ping QCCWazuh
ping: QCCWazuh: Temporary failure in name resolution
ubuntu@wazuhclient:/etc$
```

    c. We are going to use the text editor nano to edit the 'hosts' file with the command 'sudo nano hosts'

```
  GNU nano 7.2                                                    hosts
127.0.0.1 localhost
127.0.1.1 wazuhclient

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

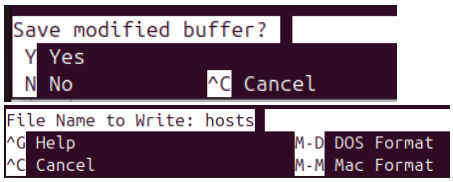
    d. Move your cusor down below '127.0.1.1 wazuhclient' and hit enter to add a new line. Then type '[WazuhServer IP address] QQCWazuh'

```
Files  nano 7.2                                                    hosts
127.0.0.1 localhost
127.0.1.1 wazuhclient
192.168.86.34 QCCWazuh

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

In my case this is '192.168.86.34 QCCWazuh'

e. Now press 'Control + X' to exit. It will ask you do you want to save the modified buffer in the lower left hand of the command terminal, press 'Y'. Then hit enter to confirm filename (overwrite). You will be dumped back out to the command line.



f. Now type 'cat hosts' to validate our changes took

```
ubuntu@wazuhclient:/etc$ cat hosts
127.0.0.1 localhost
127.0.1.1 wazuhclient
192.168.86.34 QCCWazuh

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

g. Now when we ping QCCWazuh you should see a success.

```
ubuntu@wazuhclient:/etc$ ping QCCWazuh
PING QCCWazuh (192.168.86.34) 56(84) bytes of data.
64 bytes from QCCWazuh (192.168.86.34): icmp_seq=1 ttl=64 time=0.021 ms
64 bytes from QCCWazuh (192.168.86.34): icmp_seq=2 ttl=64 time=0.028 ms
64 bytes from QCCWazuh (192.168.86.34): icmp_seq=3 ttl=64 time=0.025 ms
```

h. Later when we go to register the WazuhClient to the WazuhServer we can feed it the hostname 'QCCWazuh' instadc of an IP or FQDN. That way whenever your WazuhServer's IP changes (say when you go home), you can just go into the WazuhClient's /etc/hosts file and update the QCCWazuh to point the new WazuhServer's IP.

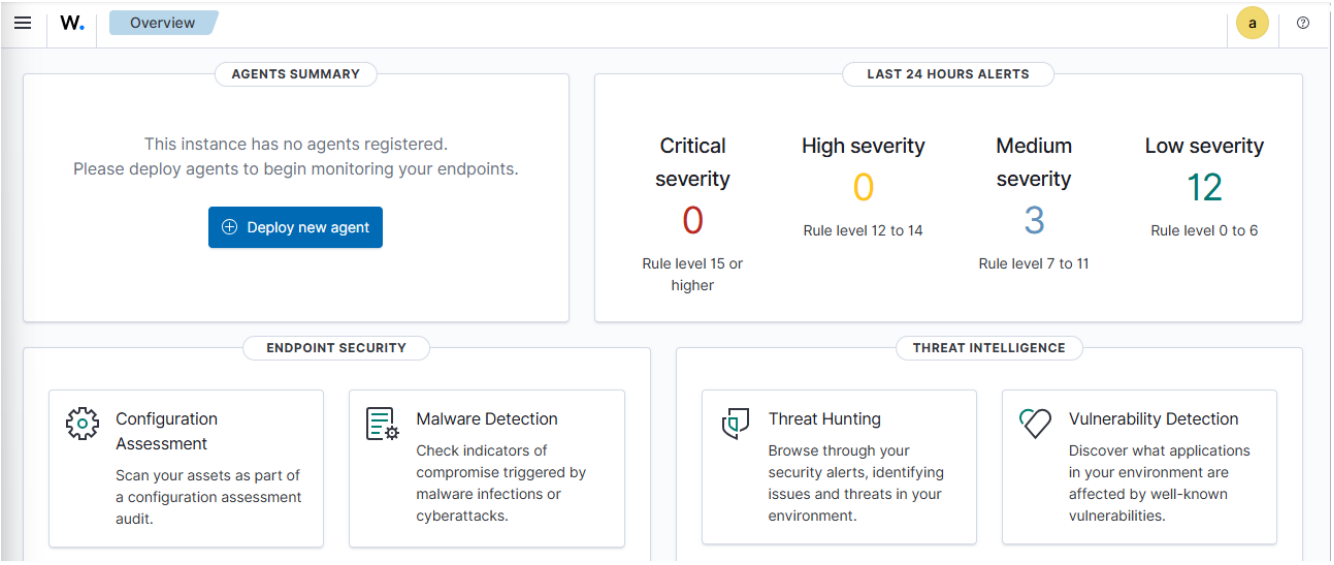
Now time to update Linux:

- At this point we want to update Ubuntu so pop open the Terminal and type 'sudo apt-get update' and press enter. Provide your password 'ubuntu' and hit enter. You instance of Ubuntu Linux will now update the local package index with the latest information of software packages available.
- Now we want to install any updated packages so type 'sudo apt-get upgrade' and watch the upgrades come in!

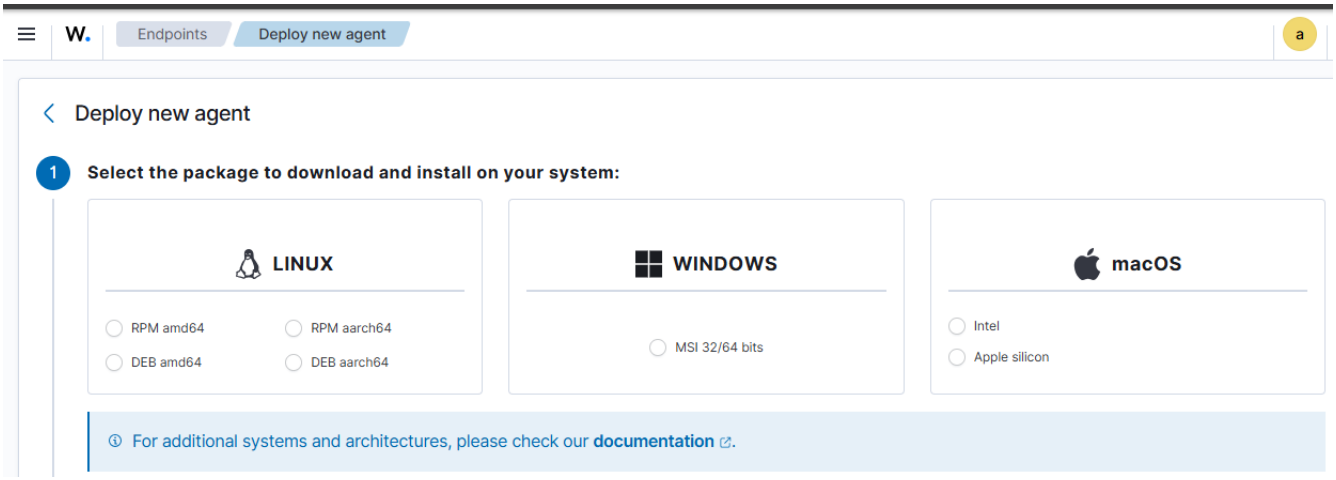
Building Wazuh Linux agent

Source: https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-linux.html

1. From your client you want to enroll navigate to the Wazuh web console: https://[Wazuh_Server_ IP]



2. Click '+ Deploy new Agent'

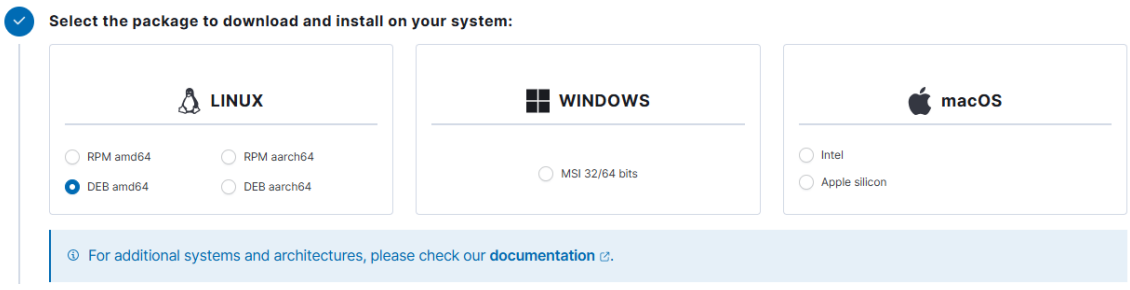


3. This is where we need to know a bit about our client and Linux architecture.

    a. First of all Ubuntu is Debian based version of Linux so we are using a 'DEB' agent. (If we were using a Red Hat version of Linux we would be interested in creating a 'RPM' agent)

    b. Next we need to understand the client CPU architecture of our host. As I am on an Intel / AMD x86-64 CPU I am going to use an 'amd64' client. (If our client was using an ARM64 CPU, then we would select 'aarch64')

    c. Putting it all together we are going to select 'DEB amd64'.



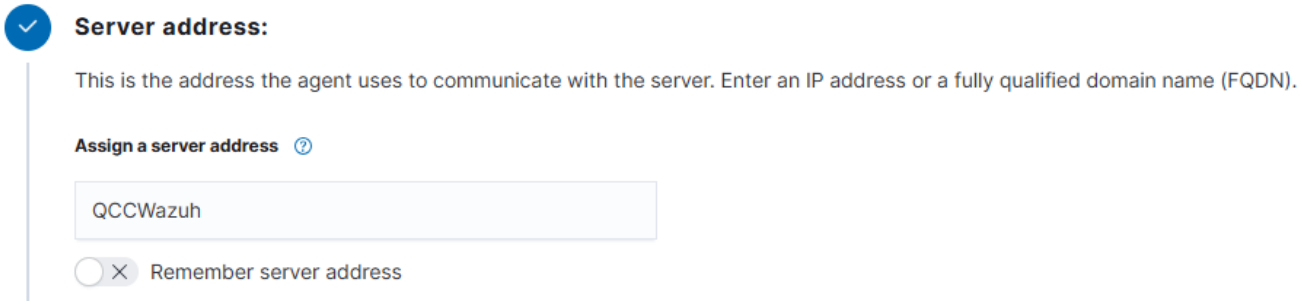
4. Next we need to specify an IP or fully qualified domain name (FQDN) to the Wazuh server. As DHCP addresses ca and will change over time, for your home lab we strong urge you to rebuild from scratch and give your Wazuh server either a DHCP reservation or static IP. (you will note for this step that my Wazuh server changed IP after I made a backup of the VM from 192.168.86.33 to 192.168.86.35, Please note that I am writing this in November 2025 and expect to do a complete rebuild before QCC 2026)

   If joining by IP:

   *Joing by IP is great for this single instance of this lab but will cause communications to fail once either the WazuhServer or WazuhClient has their DHCP reservation changed.

   ***If you made the QCCWazuh hostname entry to your WazuhClient's /etc/hosts file simply use the hostmat QCCWazuh

5. For the purpose of this lab we can keep the 'Optional settings' section default.

6. Copy the command listed in the 'Run the following commands to download and install the agent' to your text file:

   If use IP:
   wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.1-1_amd64.deb &&
   sudo WAZUH_MANAGER='192.168.86.35' dpkg -i ./wazuh-agent_4.14.1-1_amd64.deb

   If use QCCWazuh hostname:
   wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.1-1_amd64.deb &&
   sudo WAZUH_MANAGER=' QCCWazuh' dpkg -i ./wazuh-agent_4.14.1-1_amd64.deb

   You will note this is a combination of two commands '&&' together. The first command downloads the Wazuh agent and the second one uses the 'dpkg' to build / install that Wazuh agent and passes it the WAZUH_MANAGER address.

7. Copy the command listed in the 'Go to endpoints to verify the agent connection' to your text file:

   sudo systemctl daemon-reload
   sudo systemctl enable wazuh-agent

sudo systemctl start wazuh-agent

8. Execute commands from step 6 and then execute commands from step 7 into the console. (note these commands will change slightly as Wazuh is updated so it is aways best to get them directly from the Wazuh interface

9.)

```
ubuntu@wazuhclient:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.1-1_amd64.deb && sudo WAZUH_MANAGER='192.168.86.35' dpkg -i ./wazuh-agent_4.14.1-1_amd64.deb
--2025-11-22 08:13:01--  https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.1-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 3.170.131.68, 3.170.131.90, 3.170.131.63, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|3.170.131.68|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13112530 (13M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.14.1-1_amd64.deb'

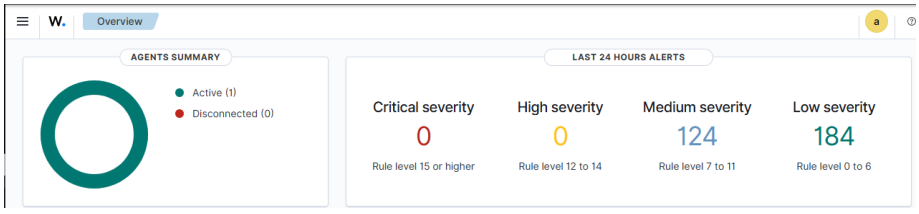
wazuh-agent_4.14.1- 100%[===================>]  12.50M  21.6MB/s    in 0.6s

2025-11-22 08:13:02 (21.6 MB/s) - 'wazuh-agent_4.14.1-1_amd64.deb' saved [13112530/13112530]

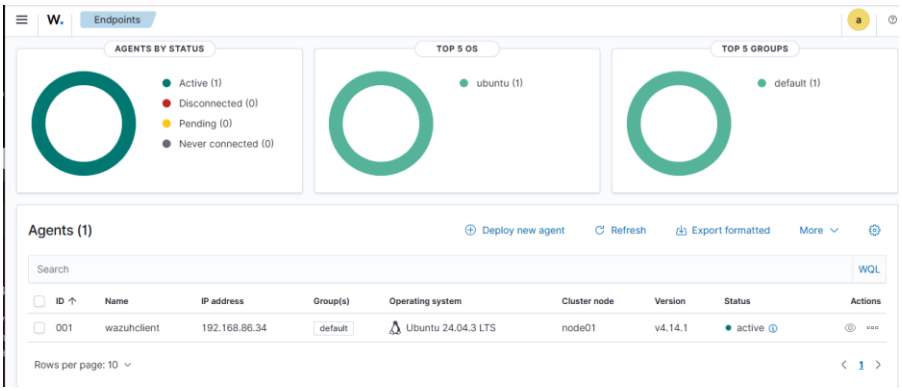
[sudo] password for ubuntu:
Selecting previously unselected package wazuh-agent.
(Reading database ... 150425 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.14.1-1_amd64.deb ...
Unpacking wazuh-agent (4.14.1-1) ...
Setting up wazuh-agent (4.14.1-1) ...
ubuntu@wazuhclient:~$ sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /usr/lib/systemd/system/wazuh-agent.service.
ubuntu@wazuhclient:~$
```

10. Now we can go back to our Wazuh admin webpage (https://[Wazuh_Server_IP]) from either the client machine or our host OS or Wazuh client. From the home sreen you will see a 'Agent Summary' with a ring, click the 'active' link to get to the 'endpoints' dashboard and you will see we have one active agent!!!

Overview dashboard:



Endpoint dashboard:

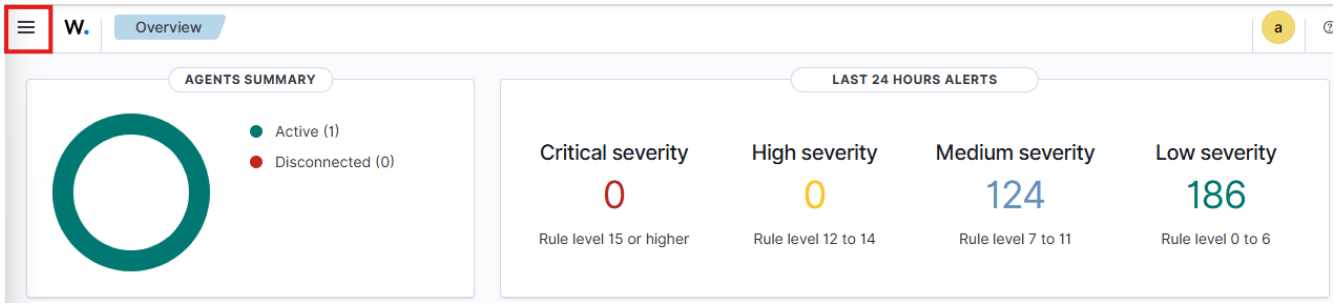


Wazuh Navigation overview!

Now we are going to take a 5 to 10 minute Talyor Swift dance break to allow our WazuhClient to report to the WazuhServer!!!

All-righty-then, I hope that break was as good for you as it was for us!

If you are not already log back into the Wazuh Web Console from either your host or the WazuhClient. On the main page (also called 'Overview') we should see some activity!



In the top left of the Wazuh interface you will see a menu button with three lines. I choose to call this the 'Cheeseburger Button' and will help us navigate though various sections of the Wuzah tool. In the main page we will see several of the most popular modules with quick links for us which you can explore at your conveniance!

- Endpoint Security
- Threat Intelligence
- Security Operations
- Cloud Security

Click the Cheeseburger button and you will see that the Wuzah Navigation Menu is devided into several section

- Home: Return to the home / overview page
- Explore: allows you to discover, dashboards, visualize, reporting, alerting, anomaly detection, maps, and notifcations.

- Endpoint Security: offers configuration assessment, malware detection, and file integrity monitoring tools.
- Threat Intelligence: Tools to help with threat hunting, vulnerability defense, and MITRE ATT&CK framework.
- Security Operations: offers tools to help with basic Governance, Risk, and Compliance tasks.
- Cloud Security: Offers several different modules to help promote cloud security.
- Agents Management: allows you to manage registered clients and groups.
- Server Management: Allows you to configure Wazuh server aspects such as rules, decoders, CDB lists, Status, clustering, statistics, logs, settings, development tools, test rulesets, and security settings.
- Indexer Management: Manage how Wazuh indexes the data coming in.
- Dashboard Management: Manage your index patterns, saved objects, Wazuh dashboard settings, and more.

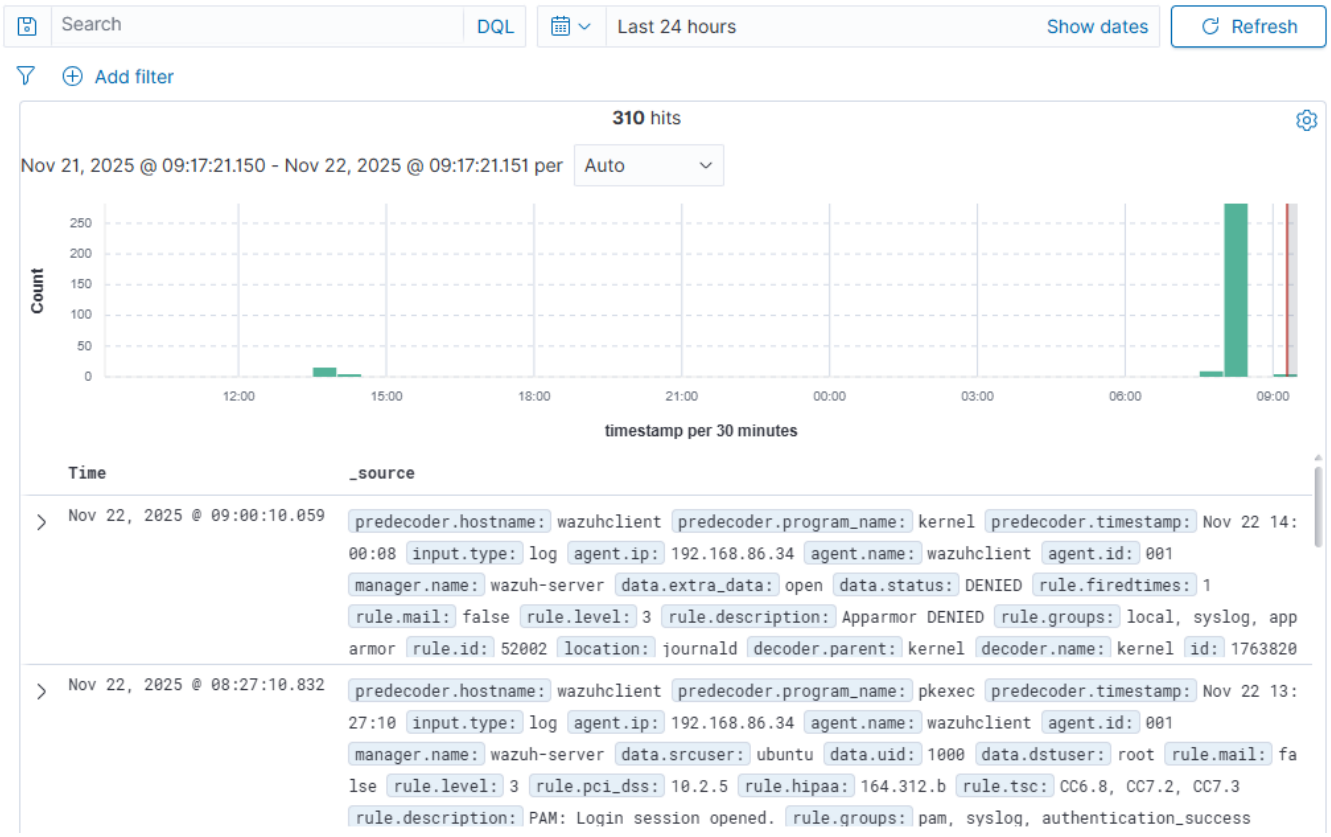
Some of these tools will have data right out of the box that we can review today. Others require you to spend some time with the friendly documentation to build out rules first for example Explorer's Dashboards & Visualizations.

I can show you the world, Shining, shimmering, splendid

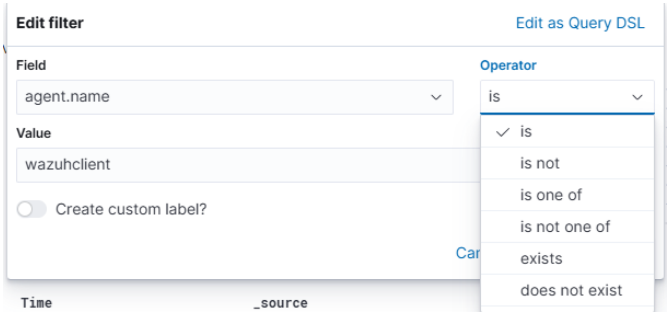
Explore:

Click on the Cheeseburger button -> Explore -> Discover

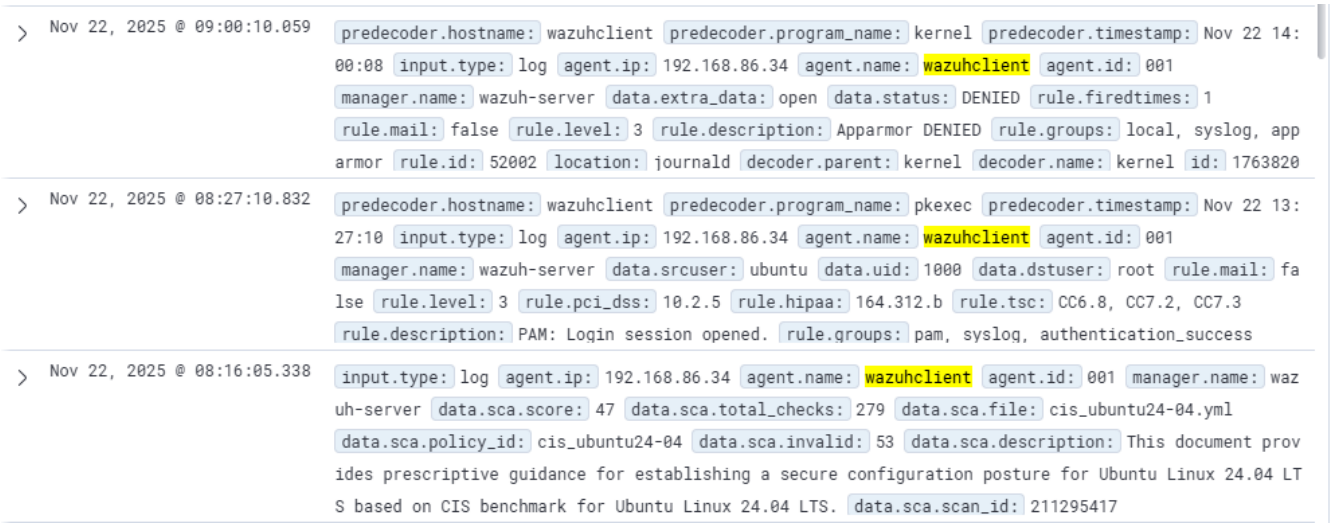
1. Click on the Cheeseburger and then ‘Explorer’ -> ‘Discover’ this will open the default ‘wazuh alerts’ view



2. This is the basic SIEM functionality of Wazuh. As most SIEMS Wazuh interprets in a [field] / [value] pair. Lets look at our logs to see.
    1. Field: agent.name / Value:wazuhclient
    2. Field: agent.ip / Value:192.168.86.34
3. On the left hand side we can see all of the parameters that we can build our filters. By default Wazuh uses DQL for it’s query language. While the scope teaching ull DQL, it is very well documented in the Wazuh documentation. For now let’s click the ‘+ Add Filter’ button just above the graph. And type type ‘agent’ in the bar and watch as it auto completes for you. Select ‘agent.name’ and select the ‘Operator’ to ‘is’. In the ‘Value’ select wazuhclient. And click save.



4. You will now see that the returned logs are those only from our wazuhclient! Additionally to make things easier for us to see, the searched for parameter is hightlighted!



5. You can then continue to add additional filters to focus in on just the logs that you care about!
6. In the top left you can bounce between different views
    - wazuh-alerts-* this is the default log view
    - wazuh-monitoring-* this view might not have much data as we just built the environment
    - wazuh-states-vulnerabilities-* we should see a small number of vulnerability alerts here
    - wazuh-statistices* this view might not have much data as we just built the environment

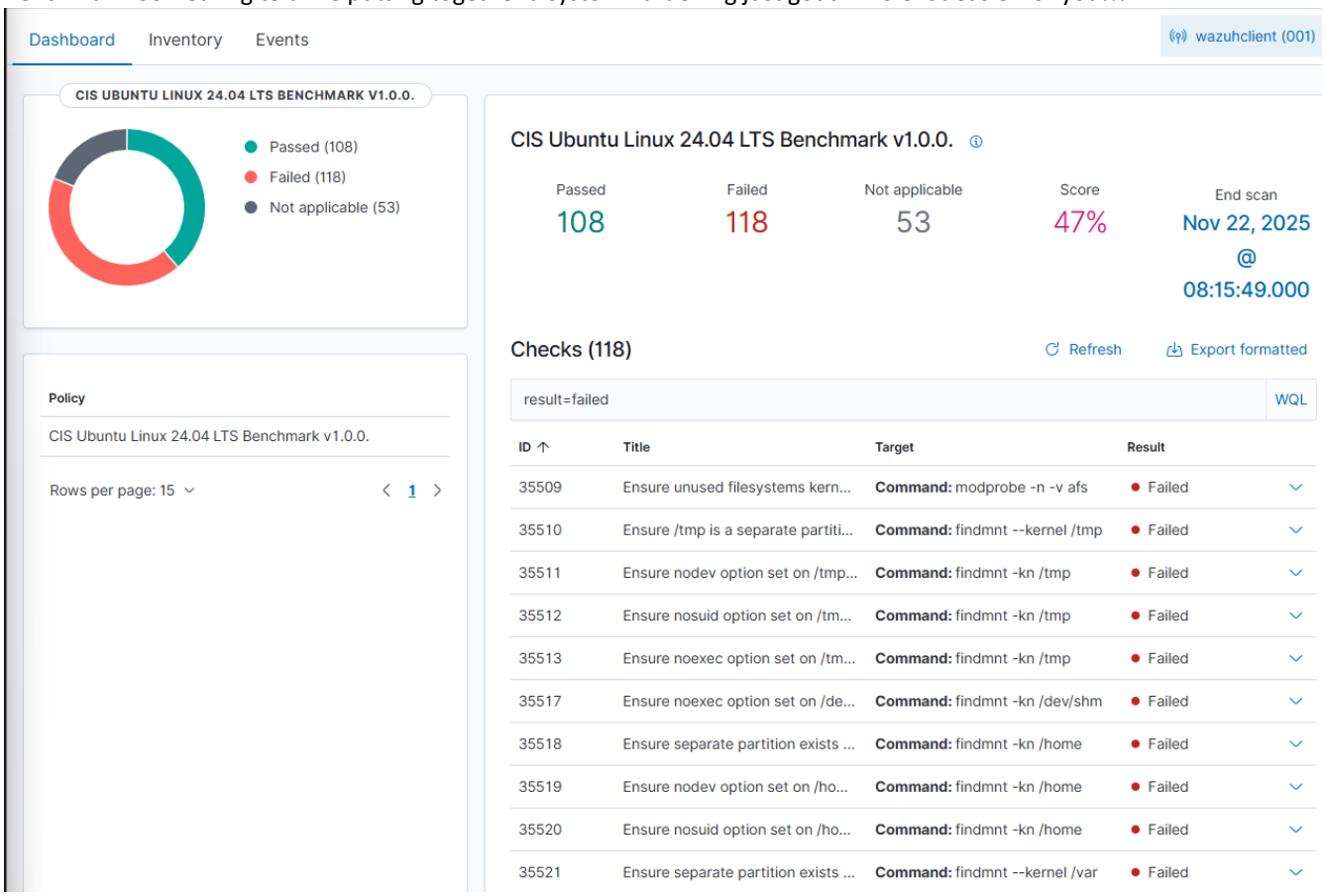
Dashboards, Visualize, Reporting, Alerting, Anomaly Detection, Maps, and Notifications all require custom setup that unfortunately we just do not have the time to cover in this lab.

Endpoint Security:

Click on the Cheeseburger button -> Endpoint Security -> Config Assessment

1. Select you wazuhClient and you should see an initial assessment dashboard. You can click on the failed number to filter just on the failed tests. Note on the left you can see that this was using the CIS Ubuntu Linux 24.04 LTS

Benchmark. Something told me putting together a system hardening just got a whole lot easier for you!!!



2. Click on the Inventory tab which is not as useful as we only have one client.
3. Click on the Events tab to look at the data in this view

Click on the Cheeseburger button -> Endpoint Security -> Malware Detection

- You can see that we do not have any detected events yet but I have not done anything on my wazuhClient that I would have expected to trigger an event

Click on the Cheeseburger button -> Endpoint Security -> File Integrity Monitoring

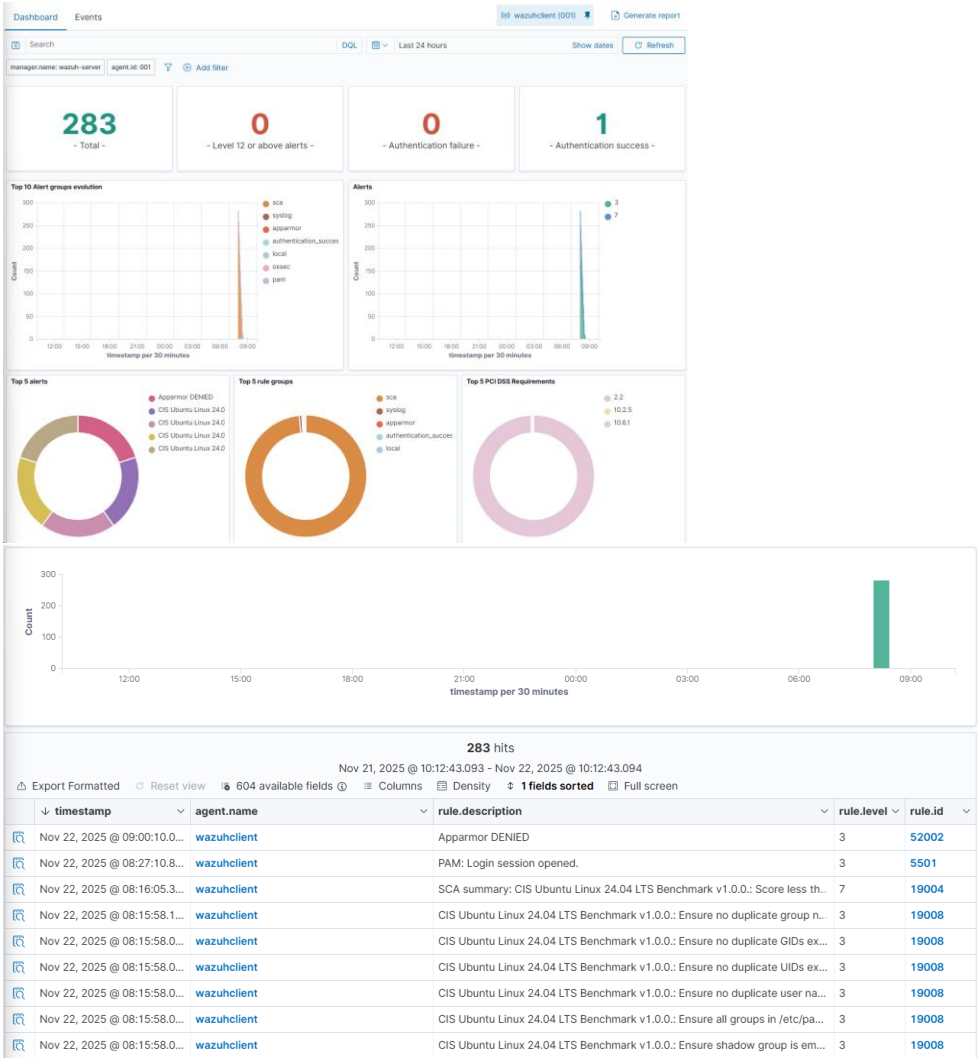
- While the dashboard will need to be set up to show any data, you should see that the Wazuh has inventoried your local disk!
- Once again I did not do enough to trigger any events.

Threat Intelligence:

(For this section it is important to remember that I did a full apt-get update && apt-get upgrade before joining my VM to my Wazuh Server)

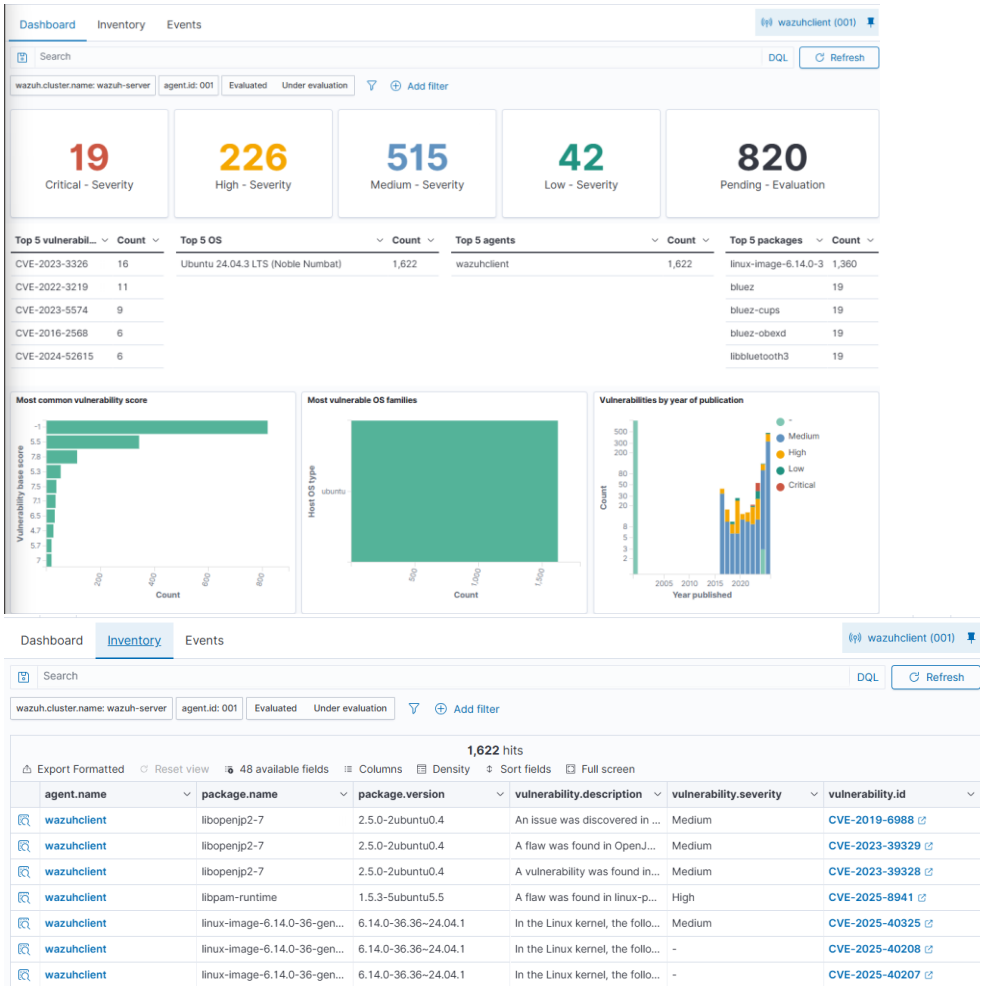
Click on the Cheeseburger button -> Threat Intelligence -> Threat Hunting

- Here we can see a basic dashboard has already populated. You can click into and out of various findings. Take a look at the ‘events’ tab to see the data in a slightly different way.



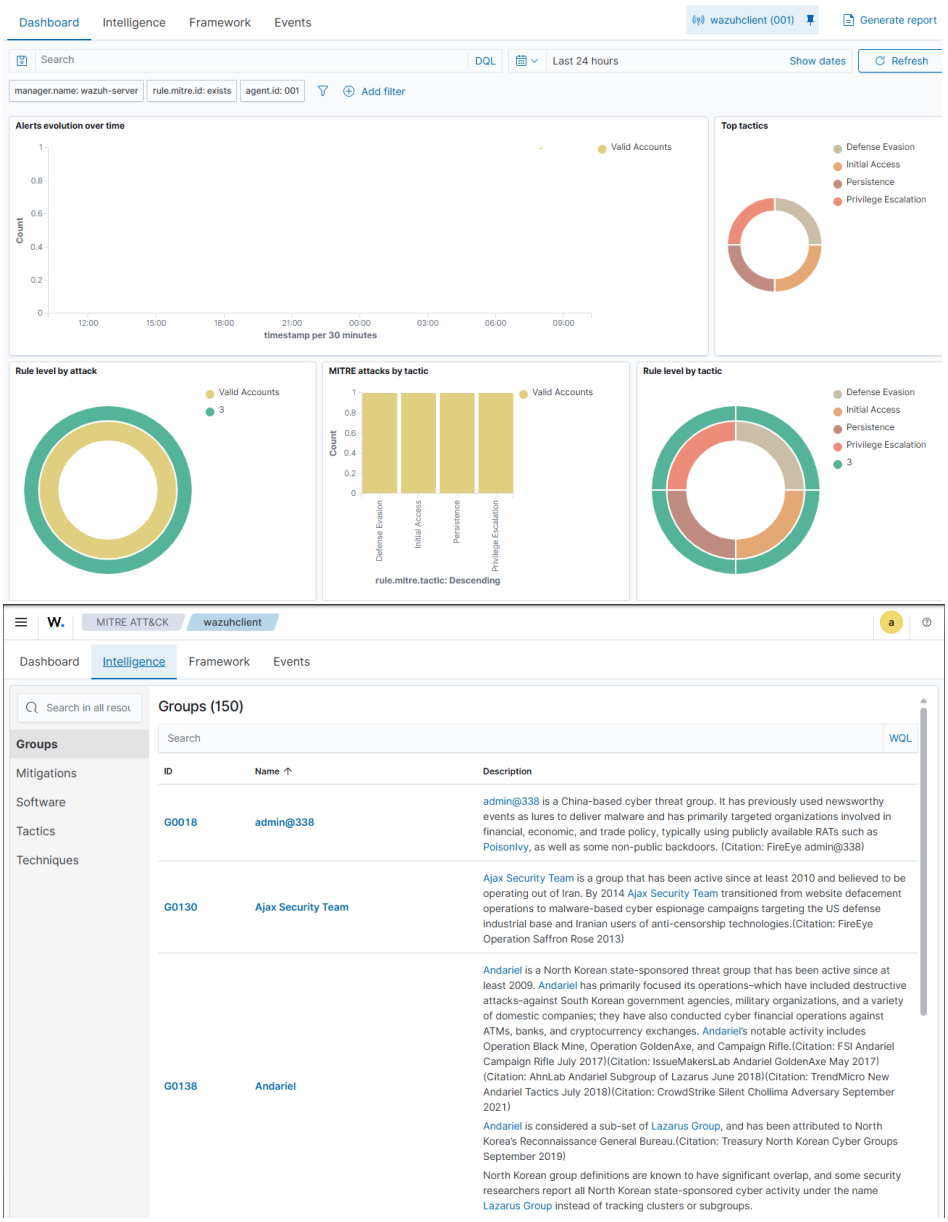
Click on the Cheeseburger button -> Threat Intelligence -> Vulnerability Detection

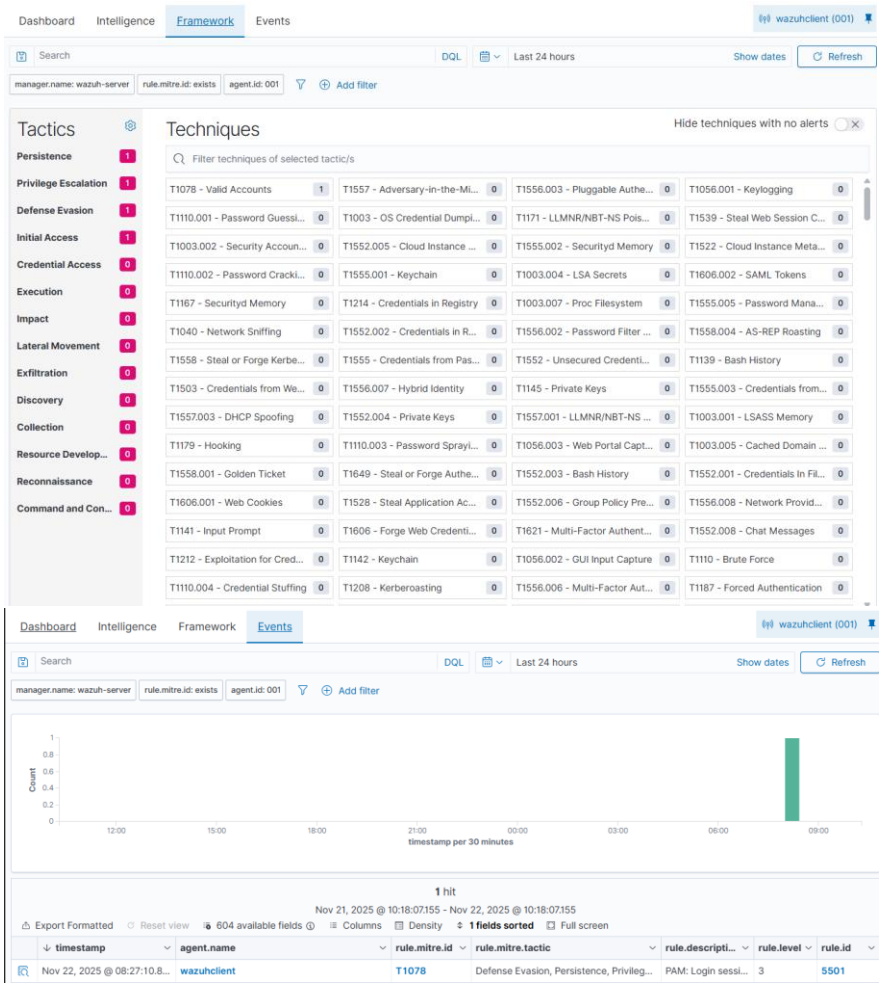
- Here we can see a basic dashboard has already populated. You can click into and out of various findings. Take a look at the ‘inventory’ and ‘events’ tab to see the data in a slightly different way. (As I had just connected my wazahClient to the server, I did not have any events to show)



Click on the Cheeseburger button -> Threat Intelligence -> MITRE ATT&CK

- Here we can see a basic dashboard has already populated. You can click into and out of various findings. Take a look at the ‘intelligence’, ‘inventory’, and ‘events’ tab to see the data in a slightly different way. (As I had just connected my wazahClient to the server, I did not have any events to show)


Dashboard
Intelligence
Framework
Events
wazuhclient (001)
Generate report
Search
DQL
Last 24 hours
Show dates
Refresh
manager.name: wazuh-server
rule.mitre.id: exists
agent.id: 001
Add filter
Alerts evolution over time
Valid Accounts
Count
timestamp per 30 minutes
Top tactics
Defense Evasion
Initial Access
Persistence
Privilege Escalation
Rule level by attack
Valid Accounts
3
MITRE attacks by tactic
Valid Accounts
Count
Defense Evasion
Initial Access
Persistence
Privilege Escalation
rule.mitre.tactic: Descending
Rule level by tactic
Defense Evasion
Initial Access
Persistence
Privilege Escalation
3
MITRE ATT&CK
wazuhclient
Dashboard
Intelligence
Framework
Events
Search in all resou
Groups
Mitigations
Software
Tactics
Techniques
Groups (150)
Search
WQL
ID
Name
Description
G0018
admin@338
admin@338 is a China-based cyber threat group. It has previously used newsworthy events as lures to deliver malware and has primarily targeted organizations involved in financial, economic, and trade policy, typically using publicly available RATs such as PoisonIvy, as well as some non-public backdoors. (Citation: FireEye admin@338)
G0130
Ajax Security Team
Ajax Security Team is a group that has been active since at least 2010 and believed to be operating out of Iran. By 2014 Ajax Security Team transitioned from website defacement operations to malware-based cyber espionage campaigns targeting the US defense industrial base and Iranian users of anti-censorship technologies.(Citation: FireEye Operation Saffron Rose 2013)
G0138
Andariel
Andariel is a North Korean state-sponsored threat group that has been active since at least 2009. Andariel has primarily focused its operations–which have included destructive attacks–against South Korean government agencies, military organizations, and a variety of domestic companies; they have also conducted cyber financial operations against ATMs, banks, and cryptocurrency exchanges. Andariel's notable activity includes Operation Black Mine, Operation GoldenAxe, and Campaign Rifle.(Citation: FSI Andariel Campaign Rifle July 2017)(Citation: IssueMakersLab Andariel GoldenAxe May 2017)(Citation: AhnLab Andariel Subgroup of Lazarus June 2018)(Citation: TrendMicro New Andariel Tactics July 2018)(Citation: CrowdStrike Silent Chollima Adversary September 2021)
Andariel is considered a sub-set of Lazarus Group, and has been attributed to North Korea's Reconnaissance General Bureau.(Citation: Treasury North Korean Cyber Groups September 2019)
North Korean group definitions are known to have significant overlap, and some security researchers report all North Korean state-sponsored cyber activity under the name Lazarus Group instead of tracking clusters or subgroups.

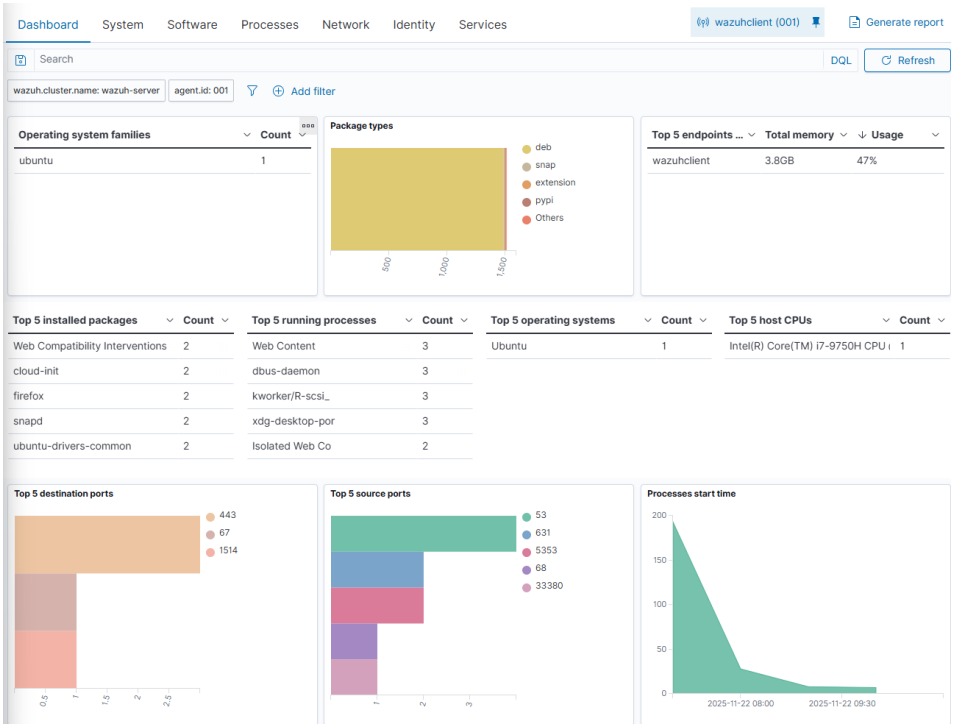


Security Operations:

- In this section Wazuh is going to take the detections we have seen in other areas and map them across several external authoritative sources.

Click on the Cheeseburger button -> Security Operations -> IT Hygiene

- We are greeted with an interesting dashboard that helps us better understand the systems connected to our Wazuh server

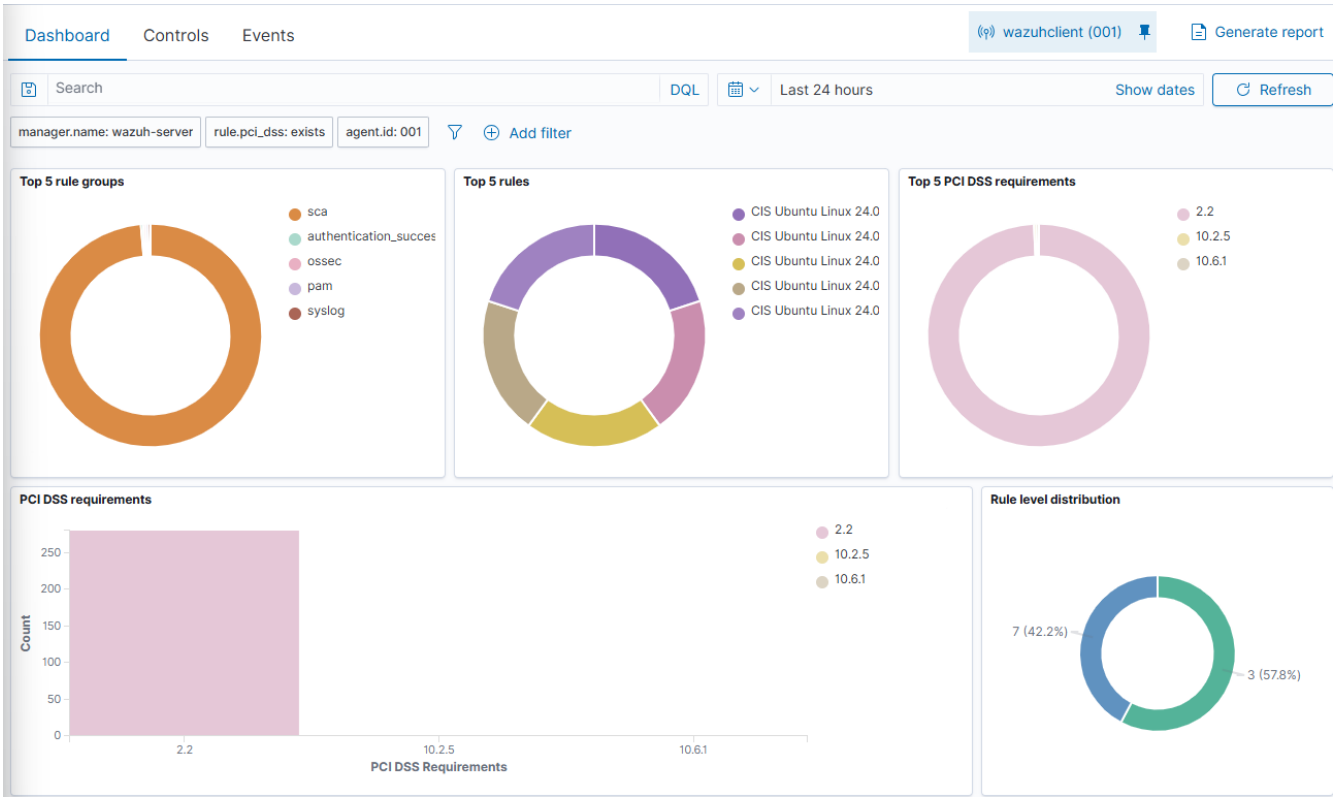


- Here we can click through the Dashboard, System, Software, Process, Network, Identity, and Services tabs.
- The ability to create a point of truth software inventory is SUPER powerful for blue-teamers.
- The fact that we can now see this detailed identity and services information is super sweat!

Click on the Cheeseburger button -> Security Operations -> PCI DSS

- PCI DSS is the Payment Card Industry Data Security Standard, a set of security standards designed to protect cardholder data and reduce fraud. It applies to any organization that stores, processes, or transmits credit card information, and compliance is mandatory for accepting card payments. Compliance ensures secure data handling, network protection, access control, and regular monitoring to safeguard payment data from breaches.

- As I have not used my Wazuh client system much, the dashboard is not that helpful (yet).

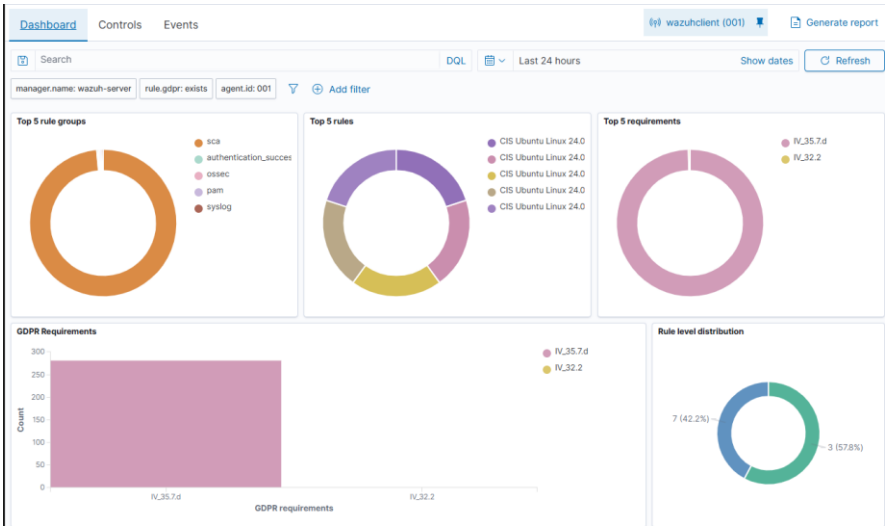


- Take a look at Controls and Events tabs
- This could help you detect and respond to PCI DSS issues in your Cardholder Data Environment ahead of an audit!

Click on the Cheeseburger button -> Security Operations -> GDPR

- The General Data Protection Regulation (GDPR) is a European Union law that protects individuals' personal data privacy and security. It establishes strict rules for how personal data is collected, processed, and stored, and grants individuals more control over their information. Key aspects include seven core principles, rights for individuals, and significant penalties for non-compliance.

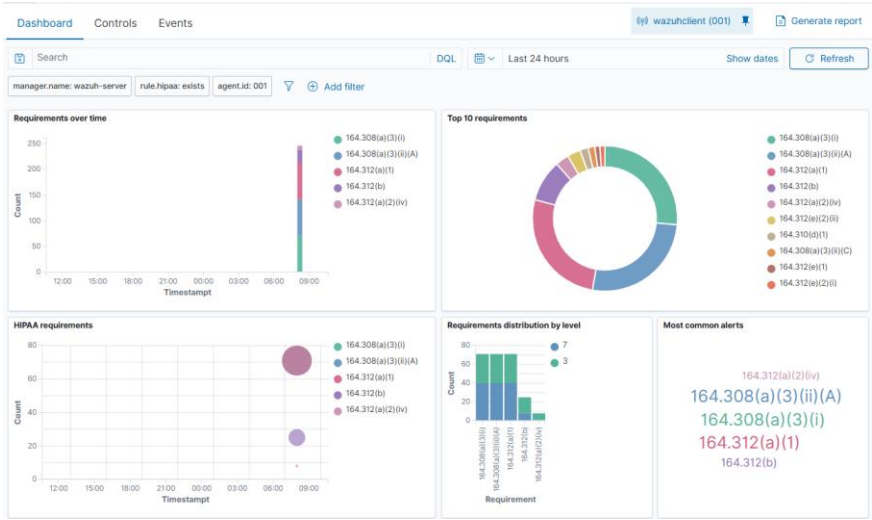
- Once again, we have a familiar Dashboard, Controls, and Events tab for this section.



Click on the Cheeseburger button -> Security Operations -> HIPPA

- HIPAA, the Health Insurance Portability and Accountability Act of 1996, is a U.S. federal law that sets national standards to protect the privacy and security of a person's health information. Its main goals are to protect patient data from unauthorized disclosure, make it easier to maintain health insurance coverage between jobs, and simplify healthcare transactions and electronic billing. The law gives individuals rights over their health information, such as the ability to get a copy of their records and request corrections

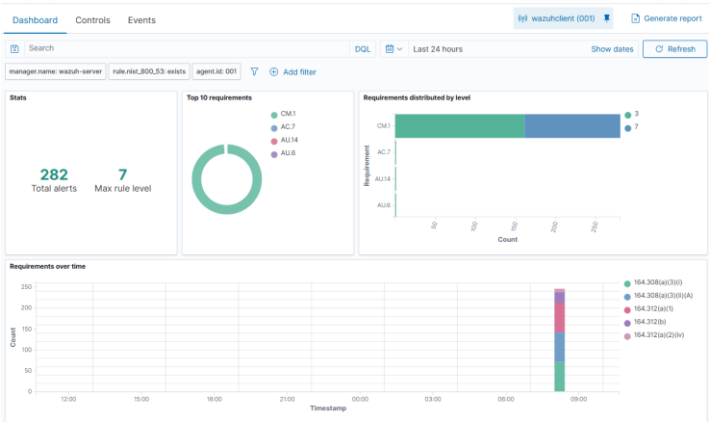
- Once again, we have a familiar Dashboard, Controls, and Events tab for this section.



Click on the Cheeseburger button -> Security Operations -> NIST 800-53 r5

Parden me friends InfoSec Brandy here with my new hit song “ NISTy girl”
“
I'm a NISTy girl in a NISTy world
Life is an audit, lets not fight it
You can scan me here, pentest me everywhere
Validation, GRC is your creation
“

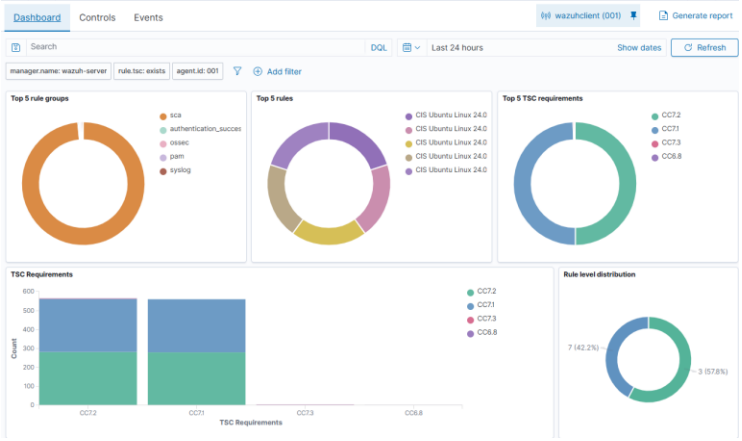
- NIST Special Publication (SP) 800-53 Revision 5 (R5) provides a robust catalog of security and privacy controls for information systems, but it is not an AI-specific framework. To address the unique cybersecurity risks of AI, NIST is developing "control overlays" that adapt the existing 800-53 controls for AI systems.
- Once again, we have a familiar Dashboard, Controls, and Events tab for this section.



Click on the Cheeseburger button -> Security Operations -> TCS

- The Trust Services Criteria (TSC) are a set of criteria developed by the AICPA to evaluate and report on the controls for a service organization's information systems. The five categories are: Security, Availability, Processing Integrity, Confidentiality, and Privacy. These criteria are used in engagements like SOC 2 audits to provide assurance on the security, availability, processing integrity, confidentiality, or privacy of an entity's information and systems

- Once again, we have a familiar Dashboard, Controls, and Events tab for this section.



Well that wraps us up with this lab! This is lab one of two having to deal with Threat Hunting. In our Threat Huning lab two we will be using X to attack our Wazuh client with know malicious activity. We will be able not only to see the logs but also hunt the threats, write queries to see what happens, and then deploy detection rules!!!

Until Next time!